Enhancing Classification Performance of ASSIRA Cats and Dogs Dataset Using Stacked Ensemble Learning Techniques

Extended Abstract:

Research Area

Deep learning models have revolutionized image classification, yet achieving state-of-the-art accuracy on benchmark datasets like ASSIRA Cats & Dogs remains a challenge due to the non-linear relationship between model complexity and performance. This study explores the potential of ensemble learning to surpass existing benchmarks by combining diverse pre-trained models, addressing the limitations of individual architectures. Recent work by Himel and Islam **[1]** further underscores this paradigm, demonstrating how hybrid architectures can optimize performance in resource-constrained environments, aligning with our focus on computational efficiency and model synergy.

Goals/Objectives

The primary objectives are to: 1) Evaluate the efficacy of ensemble methods in improving classification accuracy 2) Analyze the impact of meta-learners on ensemble performance 3) Establish a reproducible framework for optimizing model combinations

Methodology

By leveraging eight pre-trained models (e.g., Xception, VGG16, InceptionResNetV2) and testing optimizers (Adam, Adamax, RMSprop) and loss functions (binary cross-entropy, hinge), the study identifies top-performing configurations. The novelty lies in systematically applying stacking ensemble techniques with Support Vector Machine (SVM) as a meta-learner. The methodology involves rigorous experimentation with data augmentation (zoom, rotation, flipping) and hyperparameter tuning (25 epochs, batch size 32).

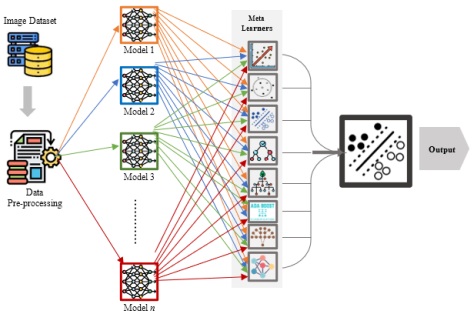


Fig 1. Proposed Ensemble method

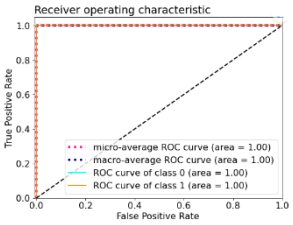


Fig 2. ROC for best ensemble combination (top 7 using SVM) found in the experiment

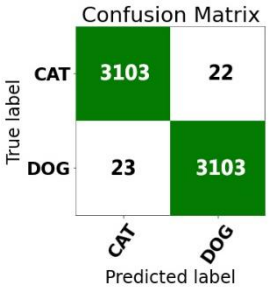


Fig 3. Confusion matrix for ensemble method using SVM on top 7 models

Results

The ensemble process combines predictions from the top seven models, achieving a record 99.28% accuracy — outperforming previous benchmarks by 0.18%. Key findings reveal that SVM as a meta-learner consistently delivers superior performance, with precision, recall, and F1-scores exceeding 99% for both classes.

Analysis

The ROC curves and confusion matrices validate the robustness of the ensemble approach, particularly in mitigating weaknesses of individual models like VGG16 and VGG19. The study highlights the non-intuitive relationship between model count and accuracy in stacking, demonstrating that strategic selection outweighs sheer quantity. While results underscore ensemble learning's effectiveness, limitations include computational overhead. Future work could explore real-time ensemble adaptation and explainability to enhance practical deployment, paving the way for advancements in ensemble-based deep learning.

Keywords: *deep learning, image classification, ensemble learning, ASSIRA Cats & Dogs.*

References

1. G.M.S. Himel, M.M. Islam, Benchmark analysis of various pre-trained deep learning models on ASSIRA cats and dogs dataset, Deleted Journal. (2025). https://doi.org/10.1007/s43995-024-00094-w.